

# Liyu Zhang

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Homepage: [zly7.github.io](https://zly7.github.io) GitHub: [github.com/zly7](https://github.com/zly7)

## EDUCATION



**Southern University of Science and Technology**

2020.09 – 2024.07

B.Eng. in Computer Science & Technology (Turing Program) | GPA 3.80/4.00, Rank 22/221



**Peking University**

2024.09 – Present

M.S. in Territorial Spatial Planning (Smart City & Big Data) | Expected graduation: 2027.07

## INTERNATIONAL INTERNSHIPS



**A\*STAR, Singapore (On-site)**

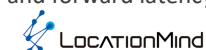
**Research Assistant**

2023.04 – 2023.09

**Research focus:** Studied the  $O(N^2)$  self-attention bottleneck of Transformer models on 10k+ long-sequence inputs.

**Sparse attention:** Implemented LSH-based sparse attention by hashing similar Query/Key vectors into shared buckets and restricting attention to local bucket interactions instead of a global scan.

**Research continuity:** This work later informed PipeFormer's topology-aware sparse attention, reducing attention computation  $\sim 200\times$  and forward latency  $\sim 200\text{ ms} \rightarrow 14.8\text{ ms}$  (first-author JPSE 2026).



**LocationMind Inc., Tokyo, Japan (On-site)**

**GPS Data Analysis Intern**

2024.05 – 2024.09

**GPS data engineering:** Processed large-scale Tokyo GPS trajectories; cleaned raw points, clustered stay locations, and built GIS/POI density/category, dwell-time, day/night, and weekday features.

**Prompt-based modeling:** Converted structured spatiotemporal features into natural-language prompts for LLM-based Home/Work/Other semantic labeling.

**Long-tail robustness:** Handled long-tail mobility patterns including shift work, frequent business travel, and multiple recurring activity locations.

## MAINLAND CHINA INTERNSHIPS



**SoTALab**

**LLM Algorithm Intern**

2026.07 – 2026.09

**SWE Agent data curation:** Curated tens of thousands of 100k–200k-token Claude Opus 4.8 / Claude Code trajectories; filtered web-answer leakage, Git-history golden patches, edit-before-read, and other invalid-agent behaviors.

**Training & inference:** Full fine-tuned Qwen-3.6-27B on 16xA100 with ms-swift; SWE-bench-Pro Claude Code improved 50.6% $\rightarrow$ 53.1%, per-task latency fell 38%, and input length fell 50%. Fine-tuned Qwen3.5-35B-A3B-Base from 35.4% $\rightarrow$ 45.3%.

**Delivery & QA:** Standardized Harbor-format delivery and defined a reproducible QA/acceptance workflow for downstream AI teams.



**ByteDance**

**LLM Algorithm Intern**

2026.04 – 2026.07

**Finance Agent:** Trained a small LLM for low-complexity Lark document operations to reduce Doubao-2.0-Pro TPM pressure during peak traffic.

**Agentic RL data:** Selected Qwen3.5-4B after tool-success/latency tests; synthesized 142 Lark CLI tool-use prompts and added 50 online tasks whose pass@8 success counts were 1–7.

**Rollout & reward:** Trained Qwen3.5-4B with GRPO (DAPO), 16k context, on 3xA100; verl full-async reduced step time from 10 $\rightarrow$ 4 min, while trajectory inspection and reward fixes improved simple document-task pass@1 from 42% $\rightarrow$ 49%.

**Mira TrustLayer:** Built an mmBERT  $\rightarrow$  Qwen3.5-4B privacy cascade and curated positive/hard-negative data using Doubao-2.0-lite screening, Doubao-2.0-pro reclassification, and human review.

**Privacy label optimization:** Fixed false positives across “Technical Solution”, financial-data, and strategy labels; at 90% recall, “Technical Solution” precision rose 70% $\rightarrow$ 78%, reaching  $\sim 85\%$  overall with keyword augmentation.



**Tencent TEG**

**LLM Evaluation Intern**

2025.03 – 2025.06

**Frontend code evaluation:** Rebuilt the manual evaluation SOP; API-generated task-level core requirements raised crowd-label consistency/accuracy from 80% to 92–94%.

**STEM data pipeline:** Built a LangChain workflow with Mistral OCR + LLM post-processing to construct hard mathematics/physics black-box evaluation sets.

**Model diagnosis:** Compared SFT/DPO checkpoints; bad-case analysis exposed undefined-variable and dead-link regressions despite visual gains, guiding data cleaning and fine-tuning.

## PUBLICATIONS & RESEARCH

**PipeFormer: A Physics-Informed Tokenized Sparse Transformer for Natural Gas Pipeline Networks | First Author · JPSE 2026 | [GitHub](#)**

**Architecture:** Designed topology-aware sparse attention, discrete tokenization, and physics-informed constraints for large pipeline networks, restricting interaction to physically relevant neighborhoods.

**Efficiency:** Reduced attention computation by  $\sim 200\times$  versus full attention and cut forward latency from  $\sim 200\text{ ms}$  to  $14.8\text{ ms}$  for transient pressure/flow prediction.

**Pipeline-Agent & PipeClaw | First Author · 2 works · JPSE 2026 | [GitHub](#)**

**Pipeline-Agent:** Built a verifiable LLM-agent workflow with explicit Plan  $\rightarrow$  Python execution  $\rightarrow$  physics/topology validation, turning natural-language requests into reproducible engineering computation.

**PipeClaw:** Added persistent operation-history memory and callable skills for PipeFormer, equipment queries, and verification; implemented a retrieve  $\rightarrow$  propose  $\rightarrow$  predict  $\rightarrow$  verify  $\rightarrow$  revise/reject dispatch loop.

**Benchmarks:** Built PipeFD-40 and PipeOpsCycle-40 for forecasting, dispatch, and long-horizon operation support.

**InstructDiff: Domain-Adaptive Data Selection via Contrastive Entropy for Efficient LLM Fine-Tuning | ACL 2026 | [GitHub](#)**

**InstructDiff method & contribution:** 10% warmup calibration + NLL filtering + contrastive-entropy ranking let 10–20% data outperform full-data SFT. Led LLaMA3-8B code experiments on 10k samples (HumanEval/MBPP/BigCodeBench), tuning selection to 20%.